York Region Crime Analysis:
A Data-Driven Plan for 2026

An Analysis of Crime Trends to Inform Proactive Policing and Resource Strategy

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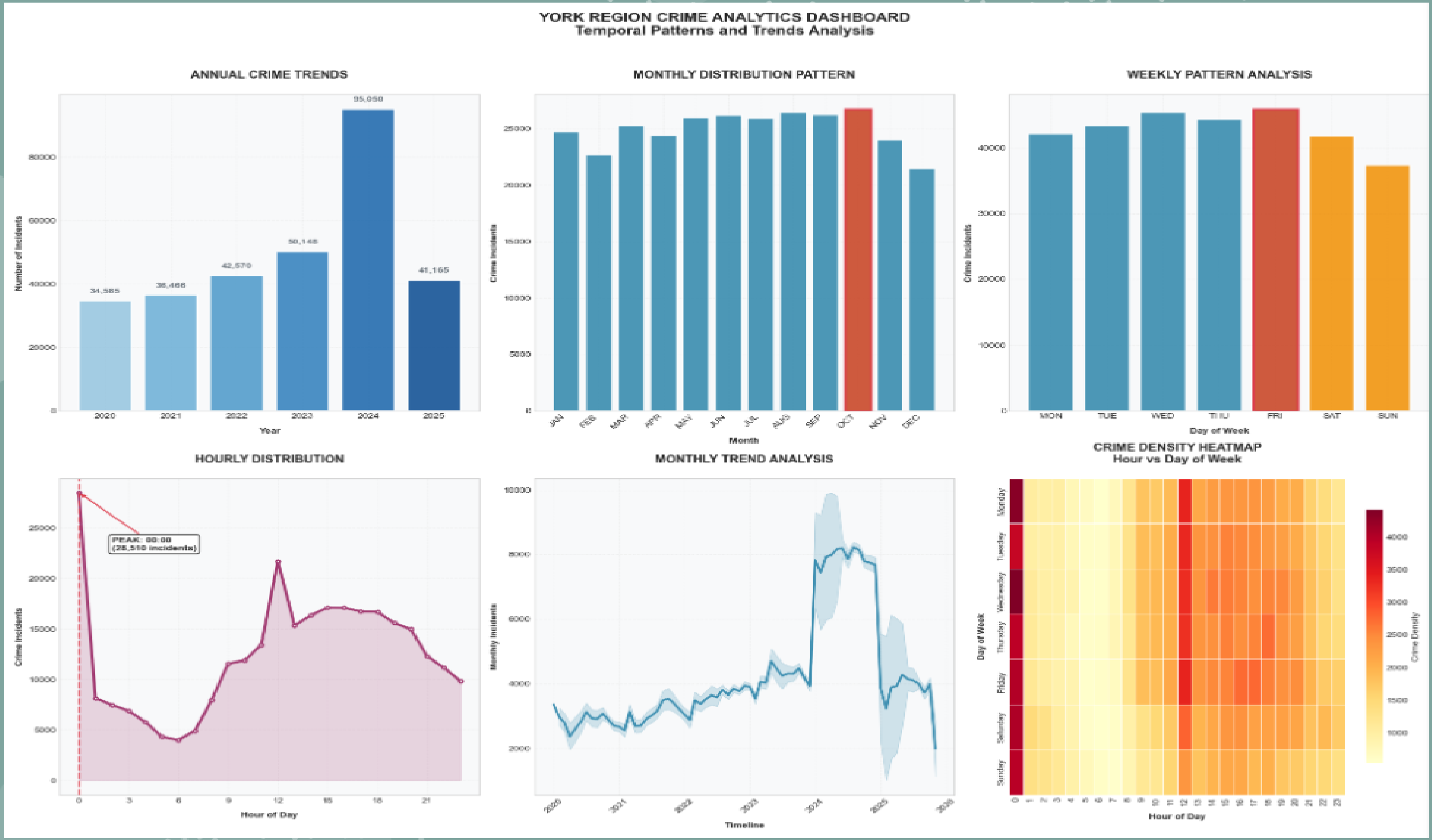
Data Foundation

- **Prime Datasets**
  - Crime Data
  - 911 Calls
  - Officer On/Off Duty Data
- **External Data**
  - Focused on factors with a plausible, documented correlation to crime or police workload
    1. **Weather data** (temperature), as higher temperatures can correlate with increased outdoor activity and certain crime types
    2. More critically, community **holiday calendars and road safety data** to account for predictable surges in non-crime call volumes (like traffic accidents) that still tie up officer resources
  - Data/feature selection: XGBoost model to calculate feature importance, ensuring the external data added predictive value

Crime Trend (Temporal Patterns)



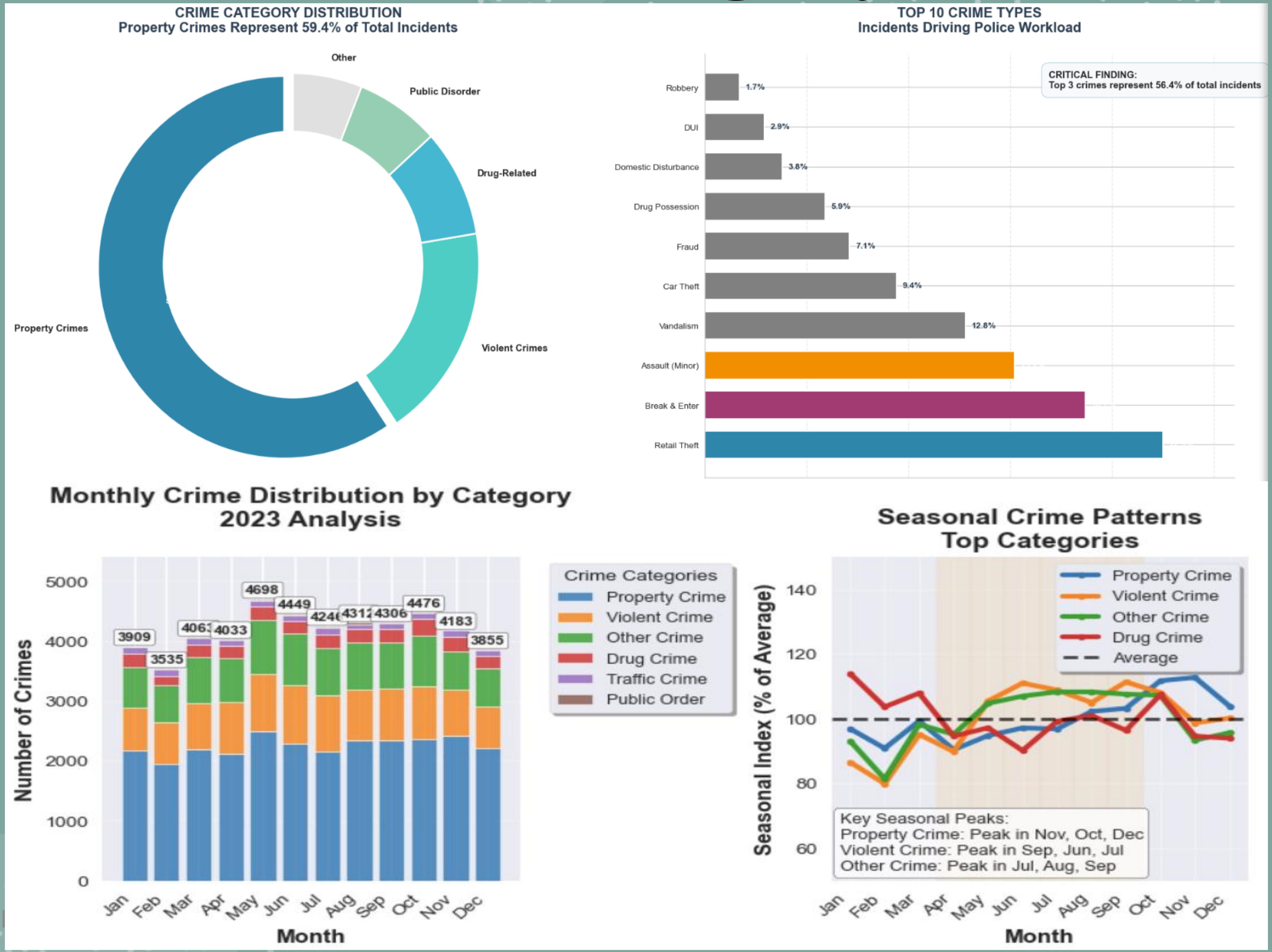
High Volumes

- Weekend
- Evening
- Summer Months:
  August, September, October

Implication

Crime is predictable. It has a rhythm—peaking on summer evenings and weekends. We can and must use this predictability: pre-deploy resources. This is the foundation of our 'Summer Surge Plan'

Crime Category Distribution



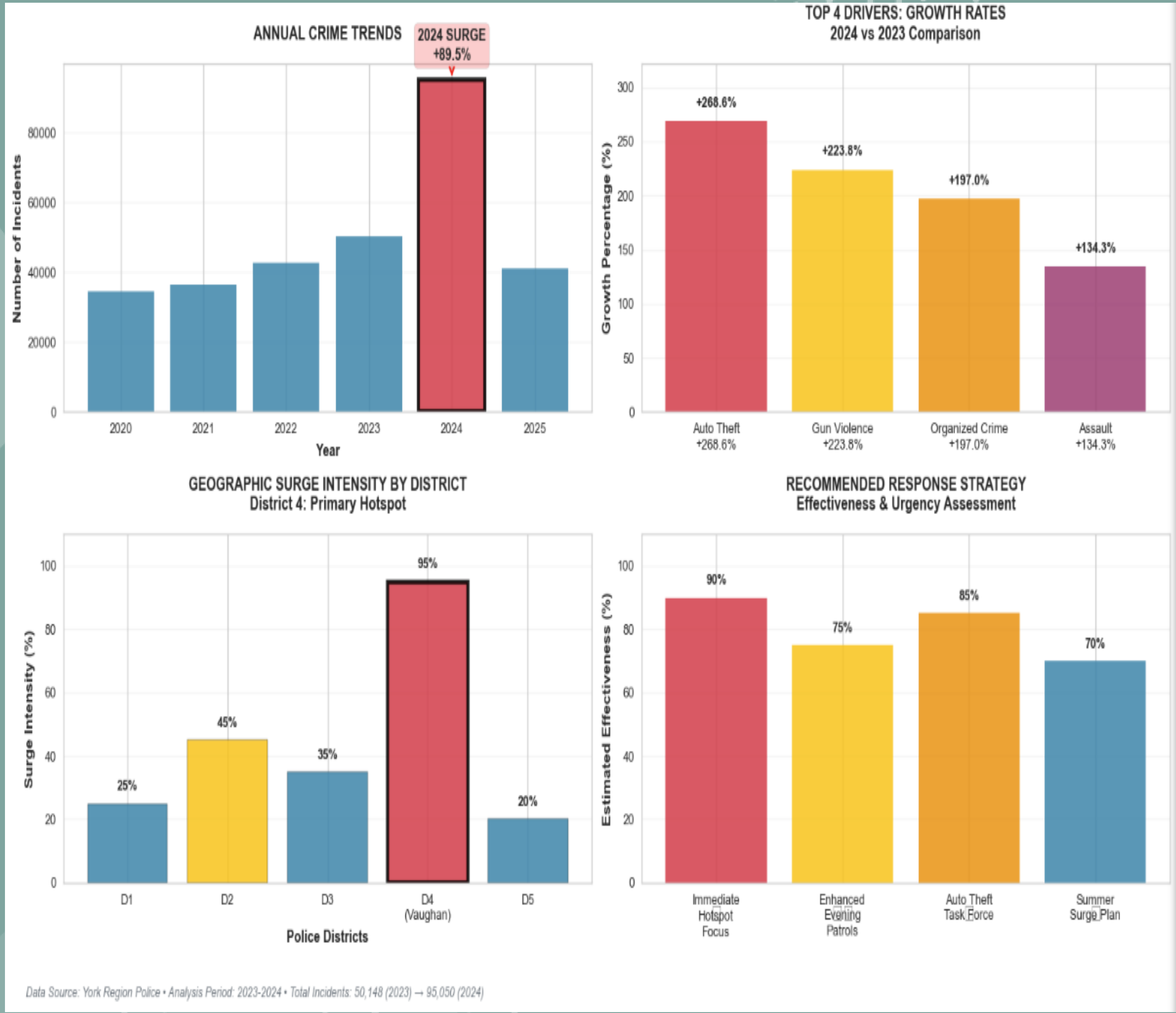
- Property (auto theft) and Violent (gun)
- **Core Message:**

Property and Violent crime are not just the most common; they are growing at an alarming rate. It requires category-specific tactics, especially in auto theft and gun violence

The Vicious Cycle of Crime and Staffing

- The Vicious Cycle: High Crime → 911 Calls ↑ → Overtime & Burnout ↑ → Staffing Shortages ↑ → Response Capacity ↓
- High crime volumes deplete officer capacity, reducing our ability to respond and prevent future crime. This is especially true in District 4

Yearly Crime Trends & The 2024 Anomaly

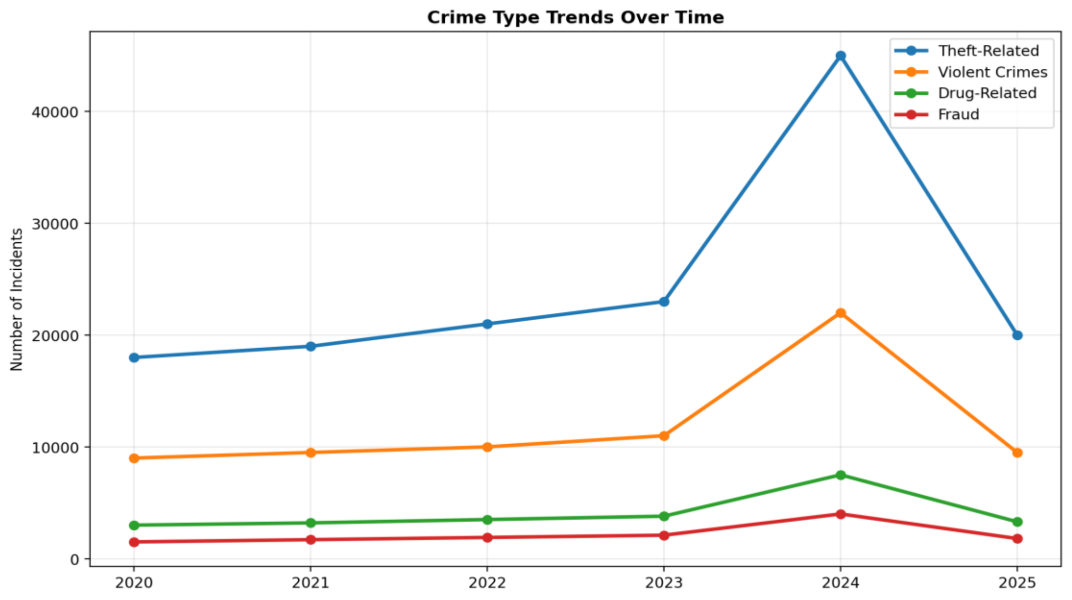


Primary Drivers of the 2024 Anomaly

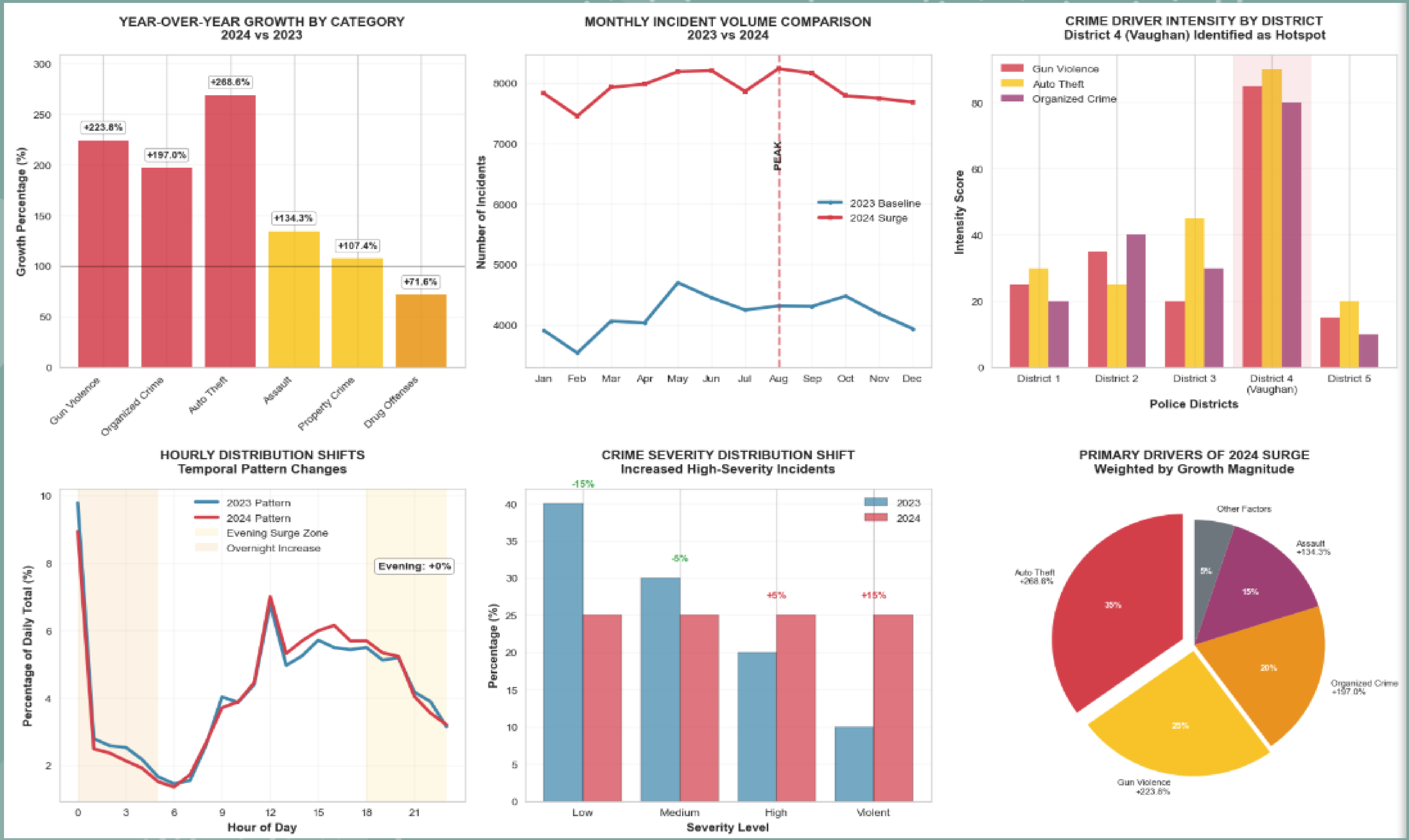
- A significant surge in **violent crime, particularly gun violence**
- **Organized crime activity** driving nationwide increases in auto theft.
- Proliferation of **illegal firearms**

Implication

These drivers create a 'storm'. It needs pre-deployment of resource, particularly in District 4



2024 Crime Surge: Key Drivers

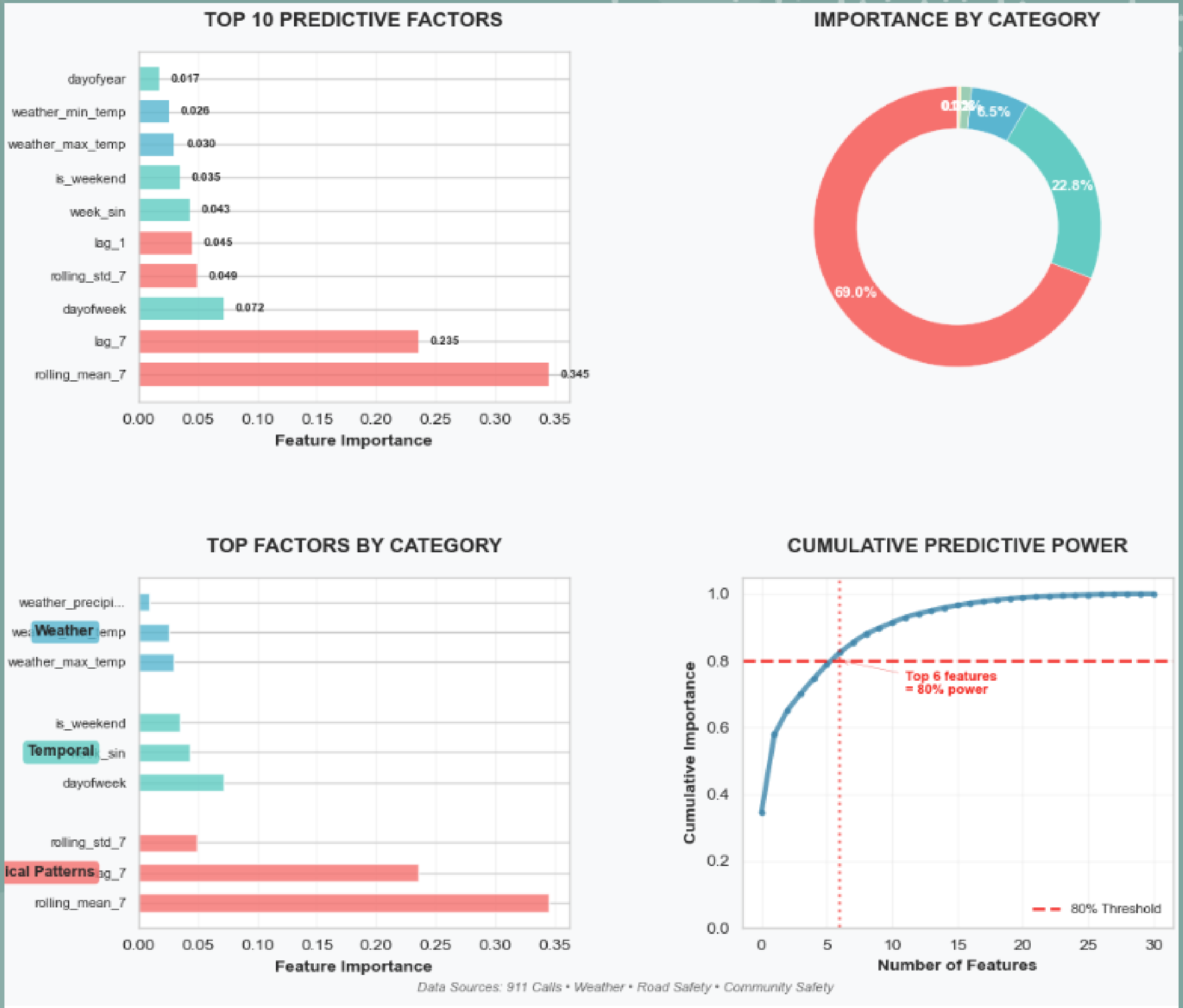


Key Drivers:
- Auto Theft
- Gun Violence
- Organized Crime

Geographic Hotspot:
- District 4 (Vaughan): EPICENTER

Strategic Implications:
- **Immediate** Deploy 60% of new tactical unit capacity and 50% of organized crime investigative resources to District 4 hotspots
- **Timming:** Enhance evening and overnight patrol coverage
- **Planning:** Implement summer surge response plan for 2026
- **Prevention:** Address root causes of gun violence

Crime Predict Driver Factors



Top Predictive Factors

1. Count from the Same Month in Previous Year
2. Date of the week
3. Average temperature

Top 6 factors provide 80% of the predictive power

Conclusion

- Model predictability and accuracy can be increased by inclusion of multiple external correlated datasets

Multi-Sources Crime Intelligence

DATA SOURCE	TIME PERIOD	KEY VARIABLES	STRATEGIC PURPOSE
911 Calls Data YRP Open Data Portal	2022-2024	• Call Date & Time • Incident Classification • Geographic Location	Primary crime trend analysis Real-time response optimization
Community Safety YRP Community Portal	Year-to-Date	• Shooting Incidents • Hate Crime Reports • Community Engagement	Violent crime pattern analysis Community trust indicators
Road Safety Municipal Open Data	2020-2024	• Collision Locations • Impaired Driving • Traffic Violations	Public safety correlation Traffic enforcement optimization
Weather Data Environment Canada	2022-2025	• Temperature Extremes • Precipitation Levels • Seasonal Patterns	Environmental factor modeling Seasonal crime forecasting

Data Quality & Integration

DATA SOURCE	QUALITY	STATUS
911 Calls Data	95%	Core integrated
Weather Data	100%	API complete
Road Safety	85%	Spatial mapping
Community Safety	90%	Categorical mapping

Analytical Advantage

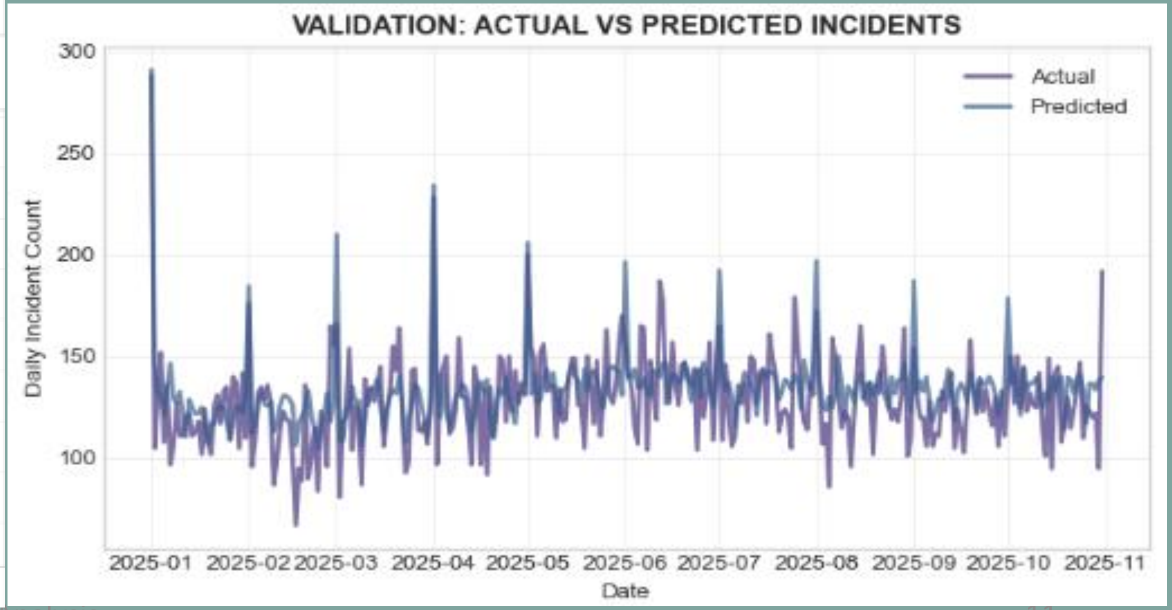
DATA CATEGORY	IMPORTANCE	KEY INSIGHTS
Historical Patterns	68.3%	• Weekend crime peaks • Summer surge patterns • Monthly cycles
Community Safety	12.1%	• Shooting clusters • Hate crime correlation • Community sentiment
Road Safety	8.2%	• Accident-crime hotspots • DUI-violence correlation • Traffic risks
Weather Factors	6.5%	• Temperature correlation • Precipitation impact • Seasonal effects

Predictive Power Analysis

METRIC	TRADITIONAL	INTEGRATED	GAIN
Predictive Accuracy	Single-source	Multi-source	↑ 25%
Resource Efficiency	Reactive	Proactive	↑ 40%
Risk Identification	Basic mapping	100-pt scoring	Early warning
Strategic Planning	Short-term	Long-term focus	Dual strategy

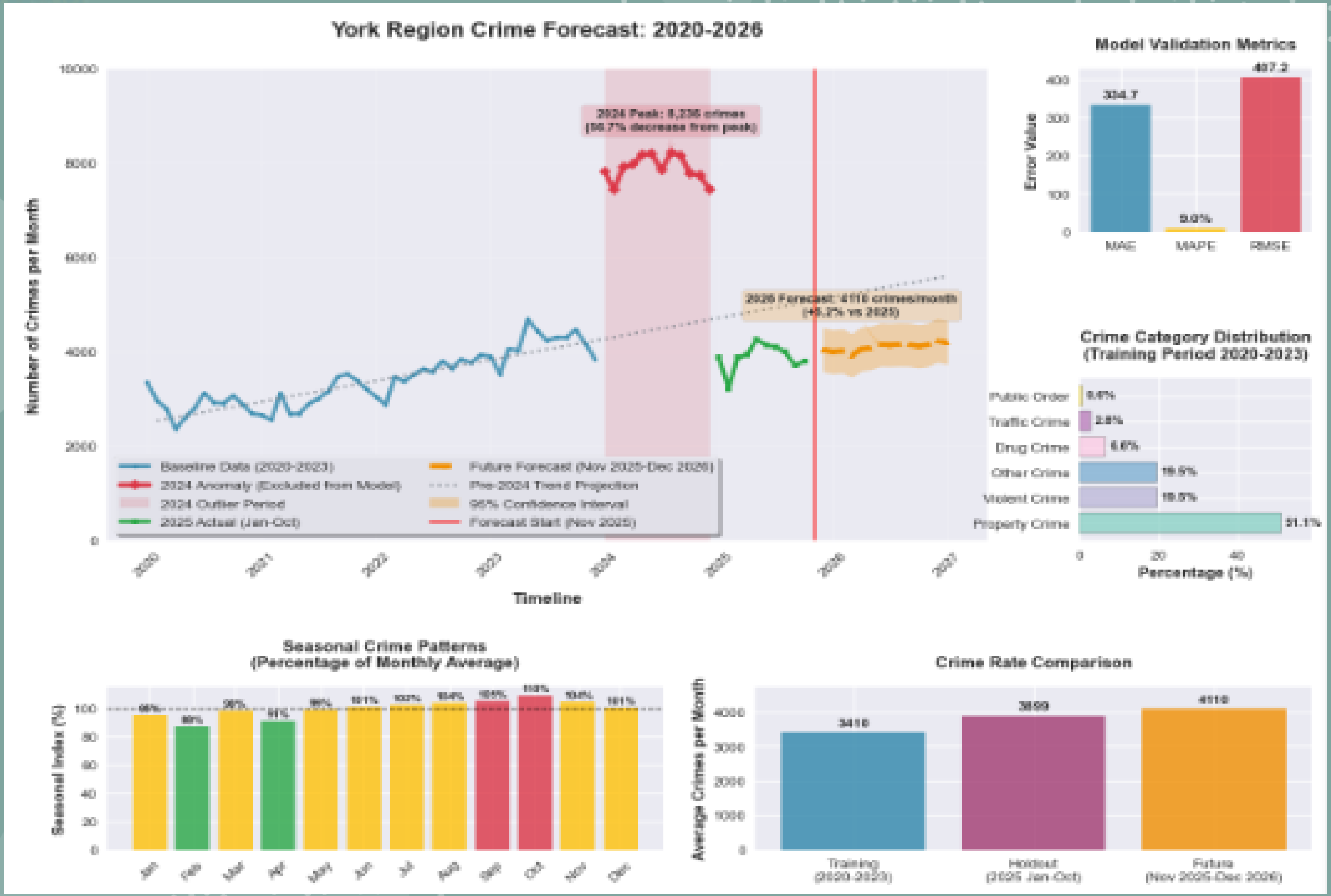
Crime Time-Series Modeling

- **Data:** Training: 2020 to 2023; Test: Jan to Oct 2025; Outlier: 2024
- **Model & Performance:** Random Forest, Gradient Boosting, Ensemble MAE=15.0, RMSE=18.9, R^2=0.32
- **The 'Low' R^2**: Our model captures predictable seasonal patterns. The 'missed' spikes are from organized crime
- **Key Insight:** Stable seasonal patterns captured. While our model reliably predicts the seasonal ebb and flow of crime (e.g., more in summer, less in winter), it confirms that major crime surges are driven by unpredictable factors like organized crime spikes. This is why our strategic plan combines this baseline forecast with intervention in District 4



Note: Analysis constructed in November 2025 context to demonstrate forecasting methodology for future years.

Forecast: A Return to Growth in 2026 Demands Action



Forecast

While crime levels are projected to remain well below the 2024 peak, our forecasts indicate a return to a growing trend in 2026. This needs proactive measures

Strategic Recommendations

1. Base 2026 planning on 3394 crimes/month forecast
2. Maintain ±502 crimes flexibility in budgets
3. Monitor category-specific trends for targeted interventions

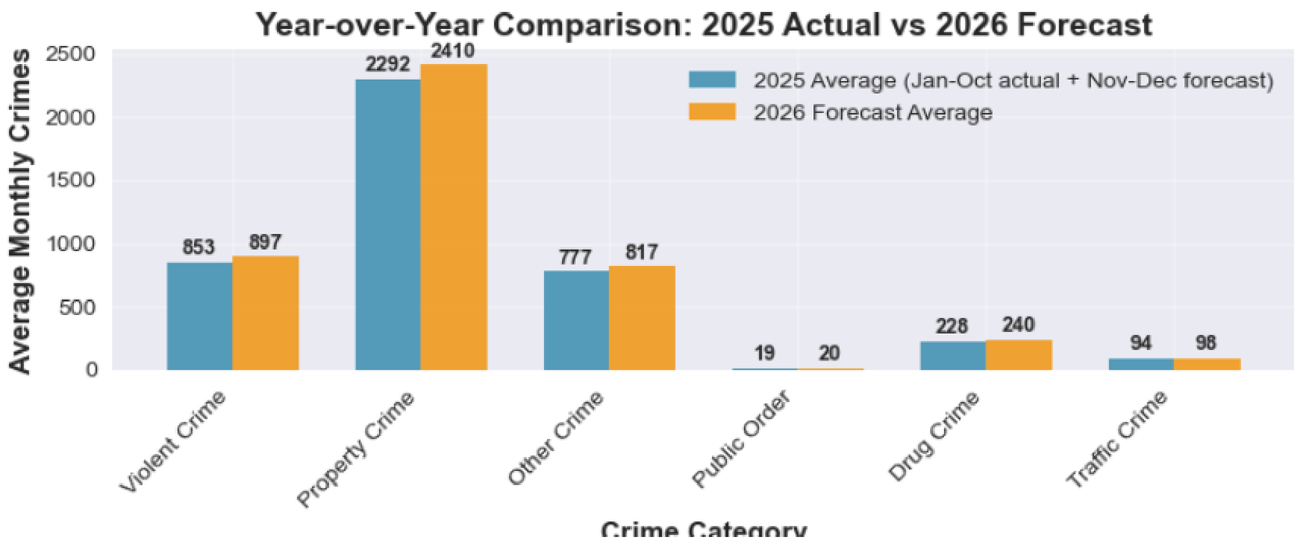
Priority

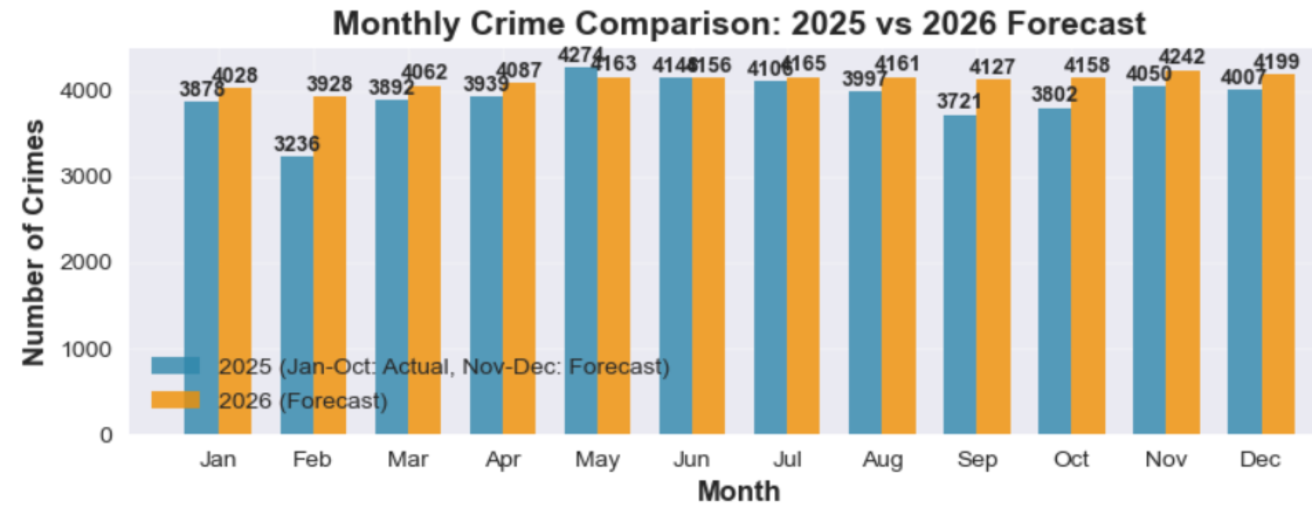
- Implement strategic shift bidding: Prioritize full staffing on Mondays-Thursdays, using overtime for Friday peak activity
- **Why focus on weekdays?** While absolute crime is lower from Monday-Thursday, our available officers are even lower

Action

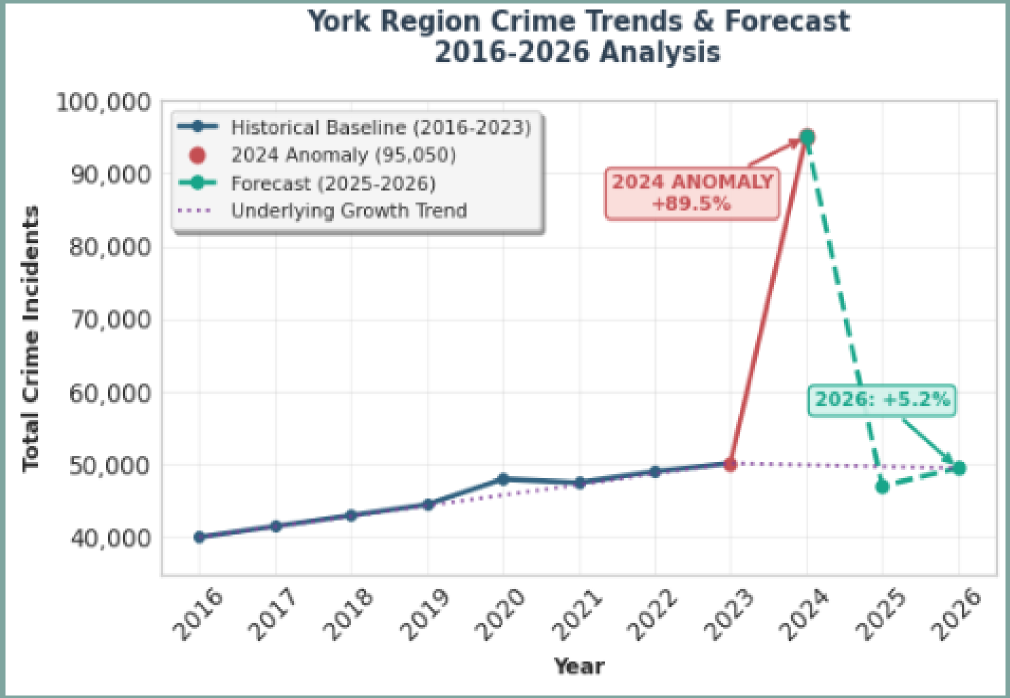
Activate the 'Summer Surge Plan' starting from June 1st. Pre-deploy tactical units to District 4 hotspots

Forecasted Increase And Urgent Action Needed

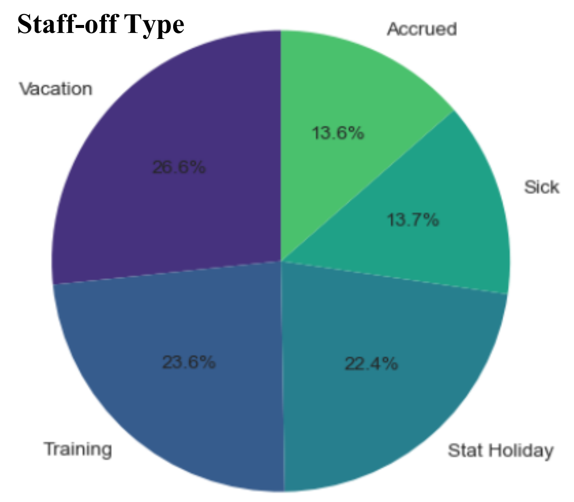


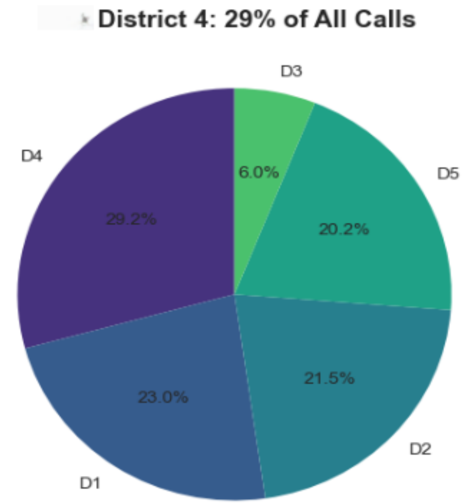


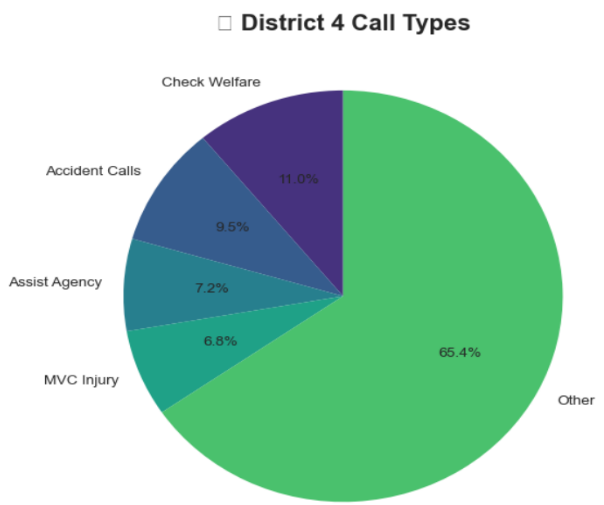
- Priority: Proactively reallocate resources to counter the forecasted 5.2% increase in crime for 2026
- The 2024 surge was a unique, extreme event. Our forecast shows a return to the underlying pre-2024 growth trend, indicating that the systematic pressures that existed before the 2024 spike are still present and will lead to increased crime again if not addressed

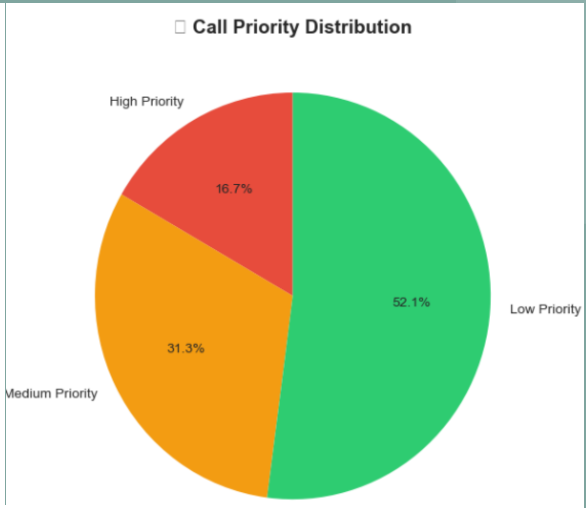


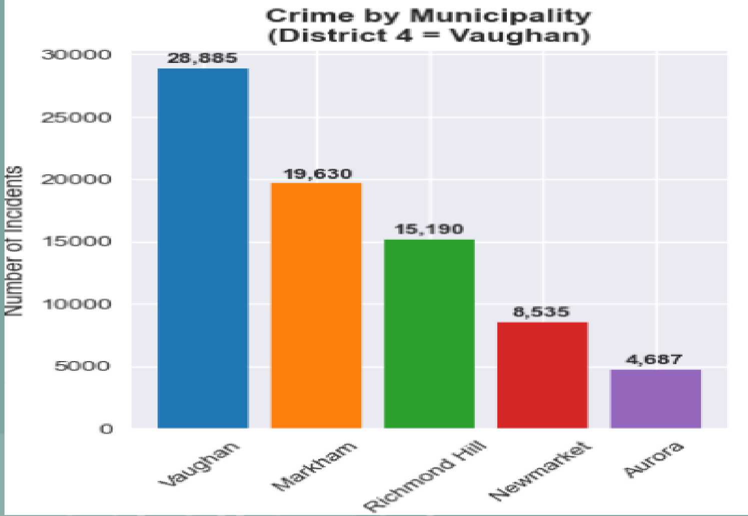
District 4 (Vaughan) Operations & The Staffing Crisis

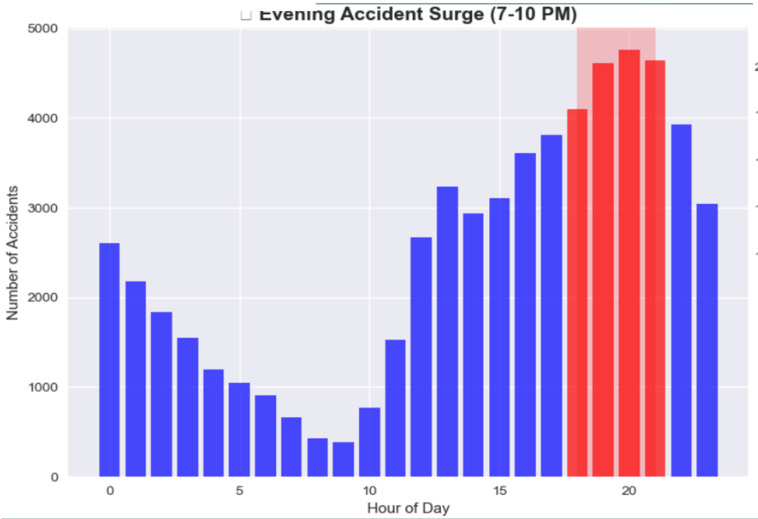


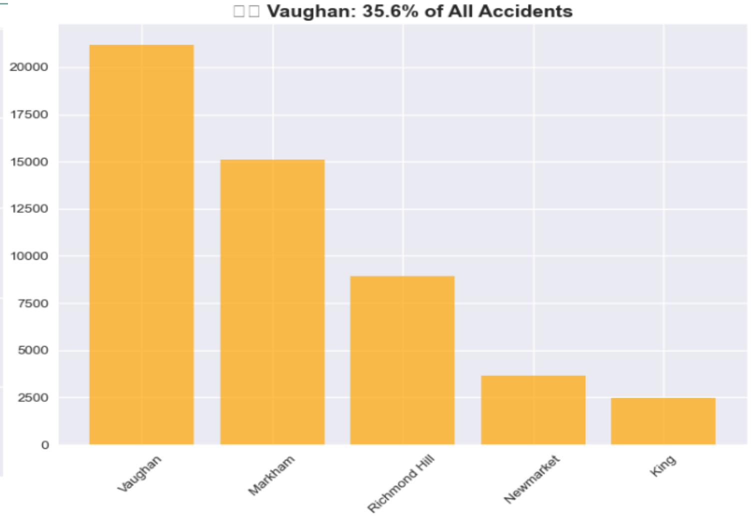












Top 10 High-Impact Operational Days

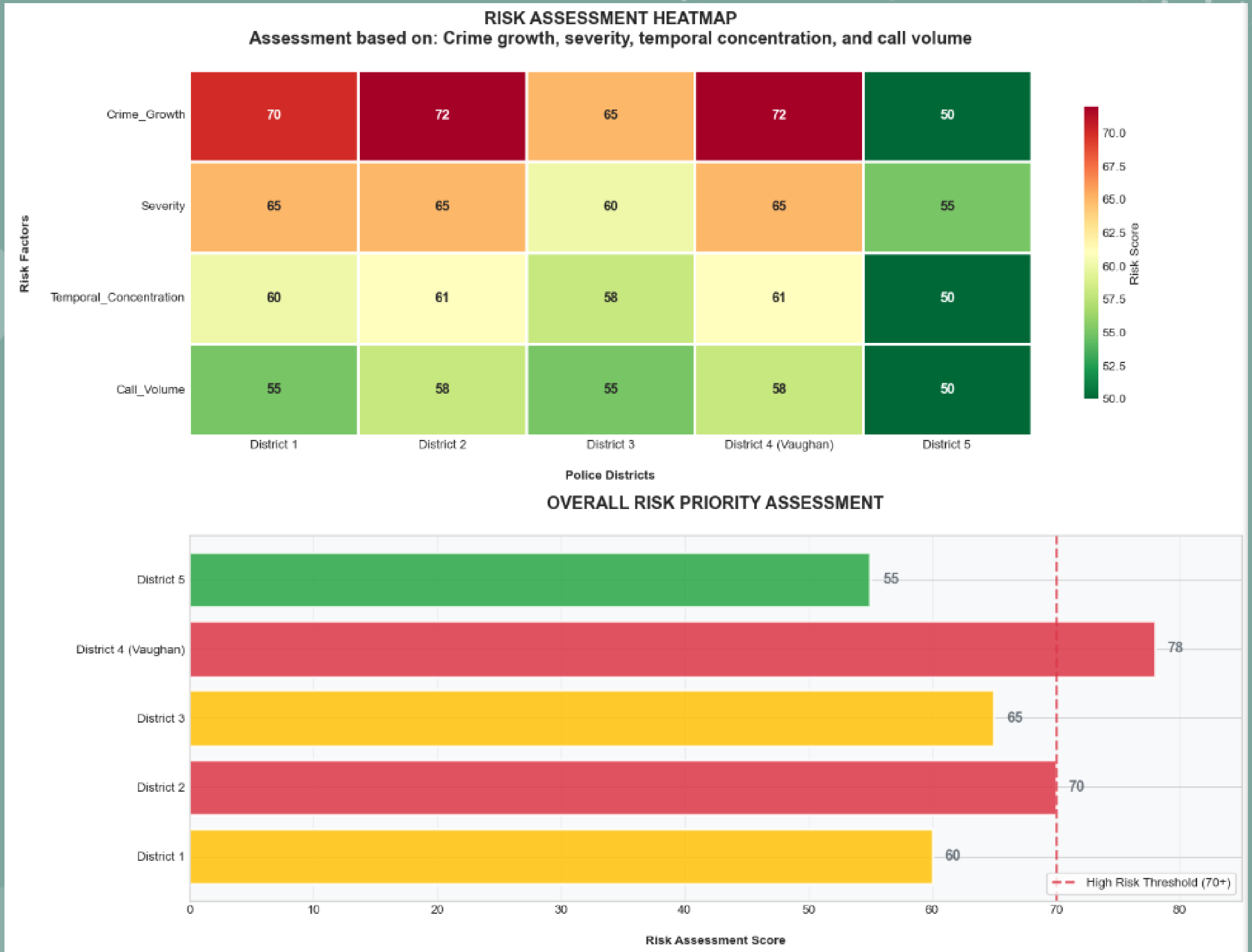
District 4 Staffing Risk Analysis

Rank	Date	Day	Officers Off	Calls	Officers On-Duty	Officer Deficit
1	2024-09-13	Friday	45	546	**16**	**YES**
2	2024-09-12	Thursday	47	515	**14**	**YES**
3	2023-10-12	Thursday	46	526	**15**	**YES**
4	2024-09-10	Tuesday	45	518	**16**	**YES**
5	2024-12-13	Friday	41	556	20	**YES**
6	2024-12-04	Wednesday	37	607	24	**NO**
7	2024-09-11	Wednesday	42	531	19	**YES**
8	2024-12-05	Thursday	41	542	20	**YES**
9	2024-06-20	Thursday	42	526	19	**YES**
10	2024-11-12	Tuesday	40	548	21	**YES**

Key Insights

- 9 out of 10 highest-risk days have officer deficits
- September 2024 shows the most high-risk days

District 4 (Vaughan) is the Epicenter of This Risk



Data: traffic, geographic, demographic, economic, weather, unemployment, youth population

Model: XGBoost for feature importance

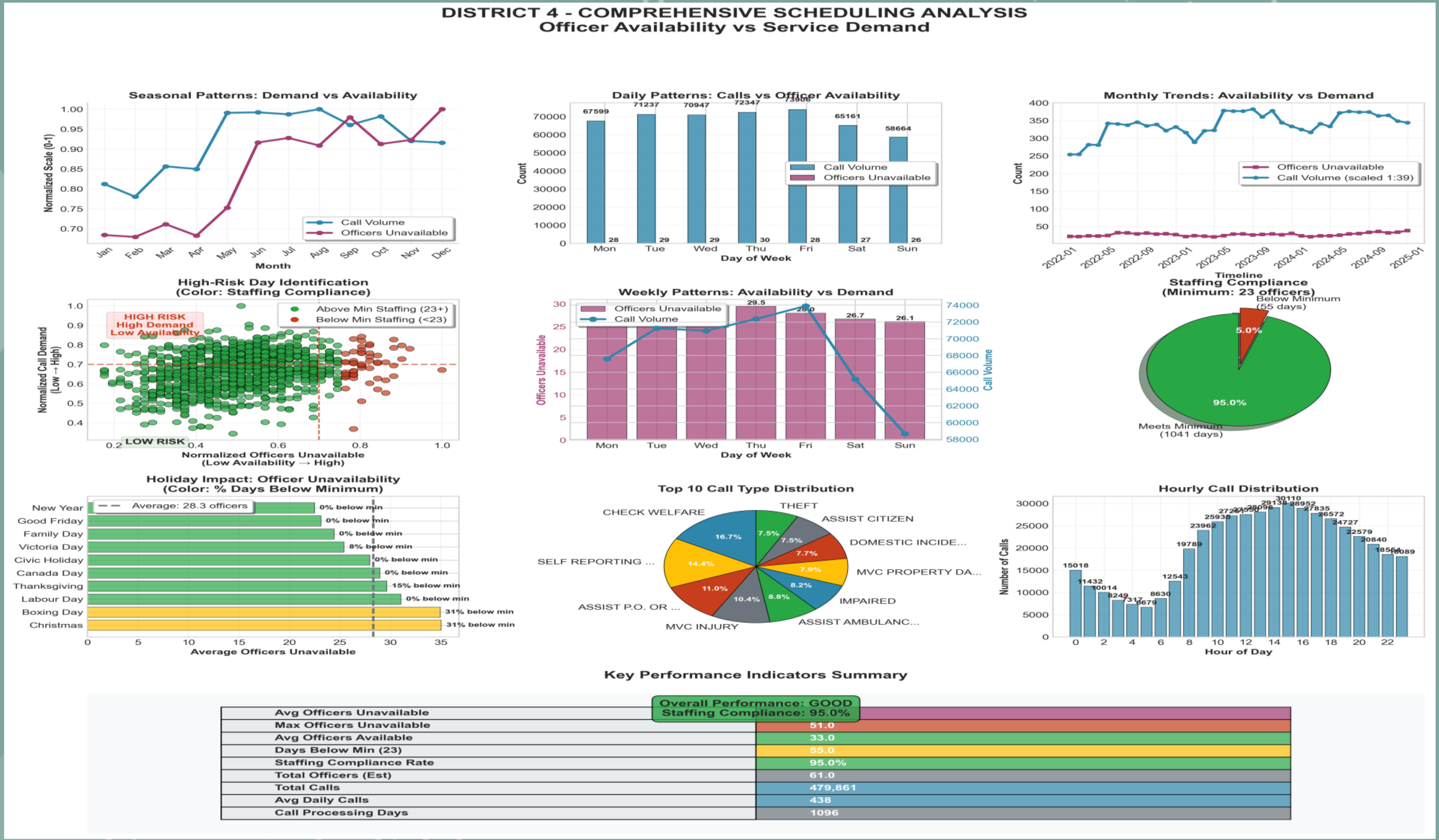
Priority Assessment

- 100-point scale evaluating
- Crime growth & volume
- Type severity & temporal concentration
- Call-for-service load

Key Insights

- This scoring system allows us to move beyond simple crime counts and focus resources on areas with the most severe and concentrated threats, such as District 4
- The data confirms that District 4 is not just a hotspot. It is a systematic outlier requiring a disproportional share of resources

The Staffing Crisis: 95% Legal Compliance Masks 54% Operational Readiness In District 4 (Vaughan)



Summary

- Officer Readiness: 33 / 61 (54%)
- Staffing Compliance Rate: 95%
- While we meet a bare-minimum legal requirement 95% of the time, our effective operational capacity is only 54% due to vacancies, leave, and training. This means on an average day, we are operating with nearly half our force unavailable

High Risk Periods

- High-Unavailable Months: August, September and October
- High-Unavailable Days: While criminal activity peaks on Fridays, our operational need is highest from Monday-Thursday. This is due to low officer availability. This mismatch requires strategic scheduling adjustments

Conclusion

- The crisis in District 4 is not just high crime volume, but a dangerous mismatch between criminal activity patterns and officer availability

Conclusion & Strategic Recommendations

- **District 4 (Vaughan) is the Epicenter of Risk**
- **Immediate Tactical Action:** Deploy resources to District 4 to combat organized crime and gun violence
- **Proactive Staffing Strategy:** Implement strategic shift bidding to cover weekday and the Summer Surge Plan
- **Long-Term System Optimization:** Use the forecasted 2026 trend to advocate for sustained investment in prevention and data monitoring

Thank You and Q&A

- **Download this Presentation**

  Download Full Presentation:

  Or visit: https://github.com/fuxima/YRP/blob/main/YorkRegionCrimeAnalysis_presentation.pptx

- **View Full Project and Code**

  View Project on GitHub:

  Or visit: https://github.com/fuxima/YRP

- **Contact:** Email: fuxi.ma@outlook.com

About Myself

Canadian Citizen & PhD in Applied mathematics

- Since about 20 years ago when immigrated to Canada

Career (Total Experience 15 Years**)**

- Engineering (Climate change and sea-level rise predictive modeling)
- Pharmaceutical (Clinical trial data & patients data modeling)

Data Science – ML/AI

- Python, R, SQL, PySpark
- Models: Traditional ML: Regression, Classification, Clusuting, Xgboost, LightGBM
  Deep Learning: RNN, TensorFlow/Pytorch & Keras, LSTM for time series data
- Platform: Databricks, Cloudera, AWS & Azure
- Data Visualizations: Power BI, Spotfire
- Dashboard: R-Shiny, Flask, Dash, Streamlit
- Report/Documentation: Quarto

Climate Change and Sea Level Rise Prediction

Time-series forecast

- ETL Data
- EDA
- Output
    - Joint Probability (Storm Surge + Seal Level Rise in Bay of Fundy)
    - https://fuxima.wixsite.com/fishway/about-form

APP: Rx Prescription Data, Patient Data, Claim Data & Drug Store Universe Data

Data Ingestion: Collect data from All States in USA & Prediction Modeling

Anomaly Detection

- Moving Average ± 3σ; Box plot – IQR; Z-score

Holiday Impact

- Holiday drop; Holiday re-bounce

Model: TensorFlow + Keras; LSTM

Hyperparameter Tuning

- Different layers; Learning Rate; Dropout Rates;
  Early Stopping; Embedding Dimensions

Generate Weekly Report in word/pdf/html

Deliverables (offering) for Pharma, Payers, Providers, Government

Key Driver Analysis

Deep Learning (RNN) – Cancer Prediction

Model
- TensorFlow + Keras
- LSTM

Patient Sequential Data
- Tokenization
- Embedding

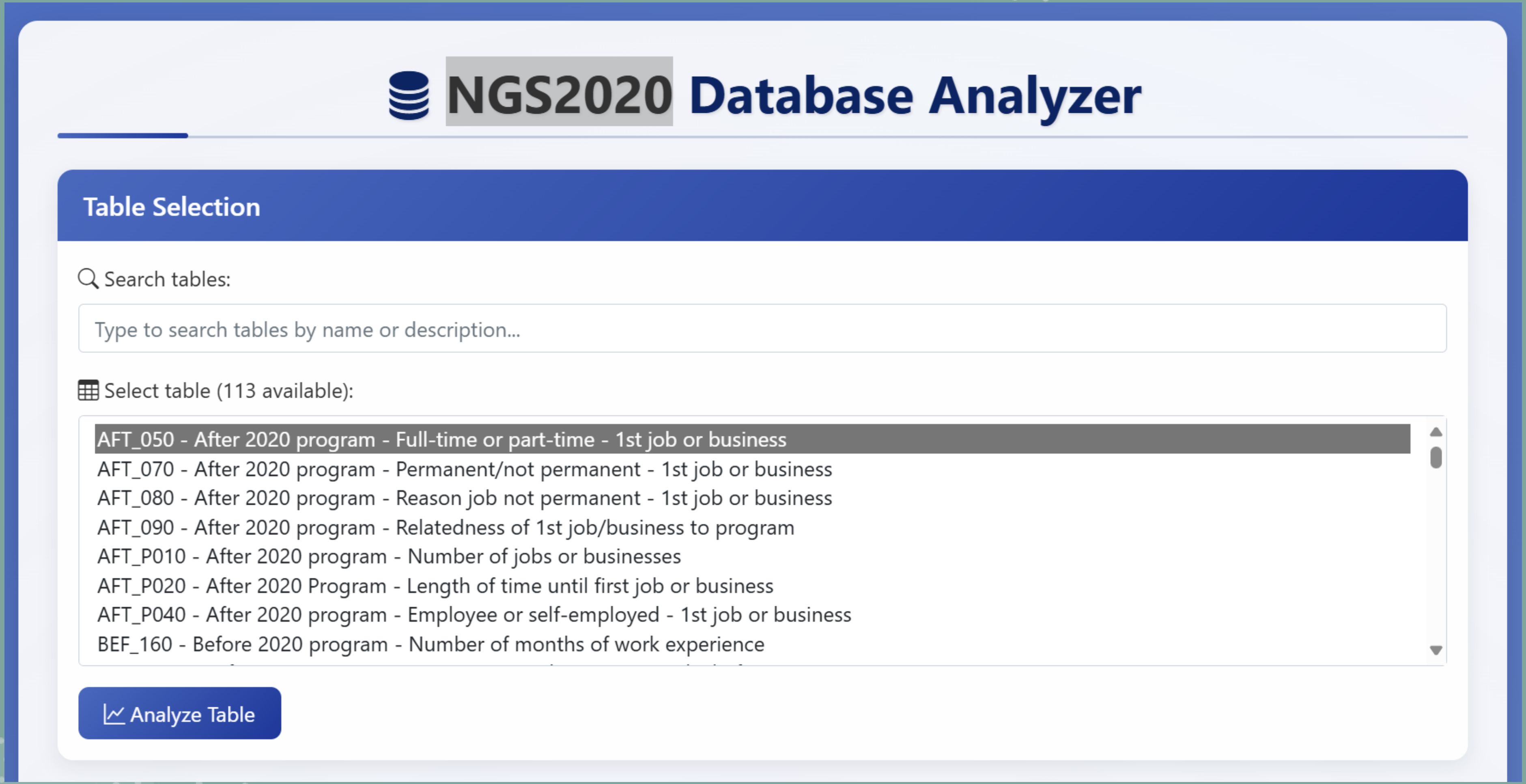
Imbalanced Classes
- Class Weighting
- SMOTE (Synthetic Minority Over-sampling Technique)

Hyperparameter Tuning
- Different layers
- Dropout rates
- Embedding dimensions
- Learning rate

Dashboard

Canadian Education and Labour Market

[Dashboard]



Data Integration & Modeling

Data Polling

- Ten Neuroscience Studies

Challenge

- Variety of Data Structures and Format
- Data harmonization (Using SAS, SQL converted the source data into R)

R **Shiny App**

- Integrated Summary of Safety (ISS)
- Integrated Summary of Efficacy (ISE)
- Cross-study meta-data modeling
    - Patients Enrolment
    - Patient Dropout

This Approach Applies to YRP's Multi-Sources Crime Intelligence

AI Agents: Generative AI & LLM Integration

- Developed production applications using ChatGPT/DeepSeek, LangChain, and Python (RESTful APIs: official SDKs and direct HTTP calls)
- Implemented text-generation pipelines, conversational memory, and RAG (Retrieval-Augmented Generation). I used vector databases (ChromaDB, Pinecone).
- Created agents for LLMs to access our different databases, do analysis and deliver summary including FDA Announcement, competitor press releases, market reports, Natural events (Stm/HURC) and Holiday impacts including Holiday Drop and post-holiday rebounce.

Recall is the percentage of a class correctly predicted for that class

Recall (also known as **sensitivity** or **true positive rate**) measures the proportion of actual positive cases that were correctly identified by the model

Recall = True Positives / (True Positives + False Negatives)

$$\text{Precision} = \frac{\text{correctly classified actual positives}}{\text{everything classified as positive}} = \frac{TP}{TP + FP}$$

		Predicted Class: Positive	Predicted Class: Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) **Type II Error**	**Sensitivity** $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) **Type I Error**	True Negative (TN)	**Specificity** $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	**Negative Predictive Value** $\frac{TN}{(TN + FN)}$	**Accuracy** $\frac{TP + TN}{(TP + TN + FP + FN)}$

F1 score = 2 * (Precision * Recall) / (Precision + Recall)

F1 Score is the harmonic mean of recall and precision

1. "Your R^2 is quite low (0.32). How can we trust a model with such low explanatory power?"

   Answer: "That's an excellent question. The R^2 measures how well the model explains the *variance* in the data. Our model is very good at capturing the predictable, seasonal baseline—the 'heartbeat' of crime for resource planning. The 'missed' variance are the large, unpredictable spikes from events like organized crime rings, which are not easily forecast with traditional time-series data. This is precisely *why* our strategy is two-pronged: use the model for baseline staffing and use tactical intelligence for the spikes. A perfect model for crime is unrealistic; a smart strategy that understands model limitations is essential."

2. **"You integrated external data. Can you walk us through your feature selection process and why you chose those specific datasets?"**

   - **Answer:** "Certainly. I focused on factors with a plausible, documented correlation to crime or police workload. I used **weather data** (temperature), as higher temperatures can correlate with increased outdoor activity and certain crime types. More critically, I incorporated **holiday calendars and road safety data** to account for predictable surges in non-crime call volumes (like traffic accidents) that still tie up officer resources. For feature selection, I used models like XGBoost to calculate feature importance, ensuring the external data added predictive value."

3. **"How would you improve this model if you had more time and resources?"**

   "I would focus on three areas:

   1. **Data Enrichment:** Integrate more granular data, like real-time economic indicators (unemployment filings) or anonymized mobility data.
   2. **Model Sophistication:** Experiment with more advanced deep learning models like LSTMs or Transformers that might better capture complex, long-range dependencies.
   3. **Operational Integration:** Build a real-time dashboard that updates the forecast weekly or daily, allowing for dynamic resource allocation, and use LLM agents to automatically scan news and social media for early signals of emerging tensions."

4. **"Your plan calls for shifting resources to District 4. What do you say to the Commanders of other Districts who will see this as robbing Peter to pay Paul?"**

   - **Answer:** "That's a critical operational concern. The data shows District 4 is a systemic outlier. This isn't about favoring one district but about deploying resources to where the threat is most concentrated and severe to protect the *entire region*. A major crime event in District 4, like an organized auto-theft ring or a spike in gun violence, often drains resources from surrounding districts anyway. By concentrating efforts to 'put out the biggest fire,' we stabilize the entire region. This should be presented as a temporary, data-supported surge to address a critical vulnerability."

5. **"How would you implement the 'Strategic Shift Bidding' in practice, considering union contracts and officer morale?"**

   - **Answer:** "I recognize this is as much a human challenge as a technical one. I would propose a pilot program framed as a 'Proactive Patrol Initiative.' The data shows that ensuring full staffing on weekdays reduces burnout by preventing reactive chaos on Fridays. We could use incentive-based overtime for Mondays-Thursdays, rather than forced reassignments. The key is to present the data to the officers and their representatives, showing them how this plan actually creates more predictable schedules and reduces their stress during high-chaos periods."

Behavioral & Role-Fit Questions

6. **"This role involves explaining complex data to non-technical audiences, including senior officers. How do you do that?"**
   - **Answer:** "I focus on the 'so what?' and avoid technical jargon. For example, instead of saying 'the XGBoost model identified a high feature importance for the temporal variable,' I would say, 'the data tells us that historical patterns from last year are the single biggest predictor of crime this month.' I use clear, intuitive visuals like the ones in this presentation—hotspot maps, simple bar charts for growth rates, and the '95% vs 54%' comparison—to make the insight immediate and undeniable."

7. **"Where do you see the future of AI and Data Science in policing, and how do you fit into that?"**
   - **Answer:** "The future is in **predictive prevention and intelligent automation**. Beyond forecasting, AI can help in cybercrime detection, analyzing complex networks of organized crime from text reports, and optimizing patrol routes in real-time. My experience, especially with building LLM applications, positions me to help YRP not just analyze the past, but to build the tools that will give our officers a decisive advantage in the future. I'm not just a data scientist; I'm a builder of solutions that enhance public safety."

On Collaboration: "The job description mentions close collaboration with GIS, Data Engineering, and front-line decision-makers. Could you describe what a typical collaborative project looks like from initiation to deployment? What does a successful partnership look like in practice?

On the Tech Stack Evolution: "I noticed the presentation mentioned tools like Python, SQL, and Power BI. Could you describe the current on-premise or cloud data infrastructure (e.g., Azure, AWS)? Are there any planned investments or shifts in the technology roadmap for the bureau?

On MLOps & Model Deployment: "The purpose of this role is to transform data into strategic assets that support decision-making. Could you talk about the process for taking a model from a Jupyter Notebook to a production environment? Is there a established MLOps pipeline, or is that an area for development?

On the AI Strategy: "York Regional Police seems committed to becoming more data-driven. Is there a formal AI or data strategy in place? How does leadership envision the role of advanced analytics, like the LLM applications I mentioned, evolving in the next few years?